



Inside Stars : Modelling convective boundary mixing

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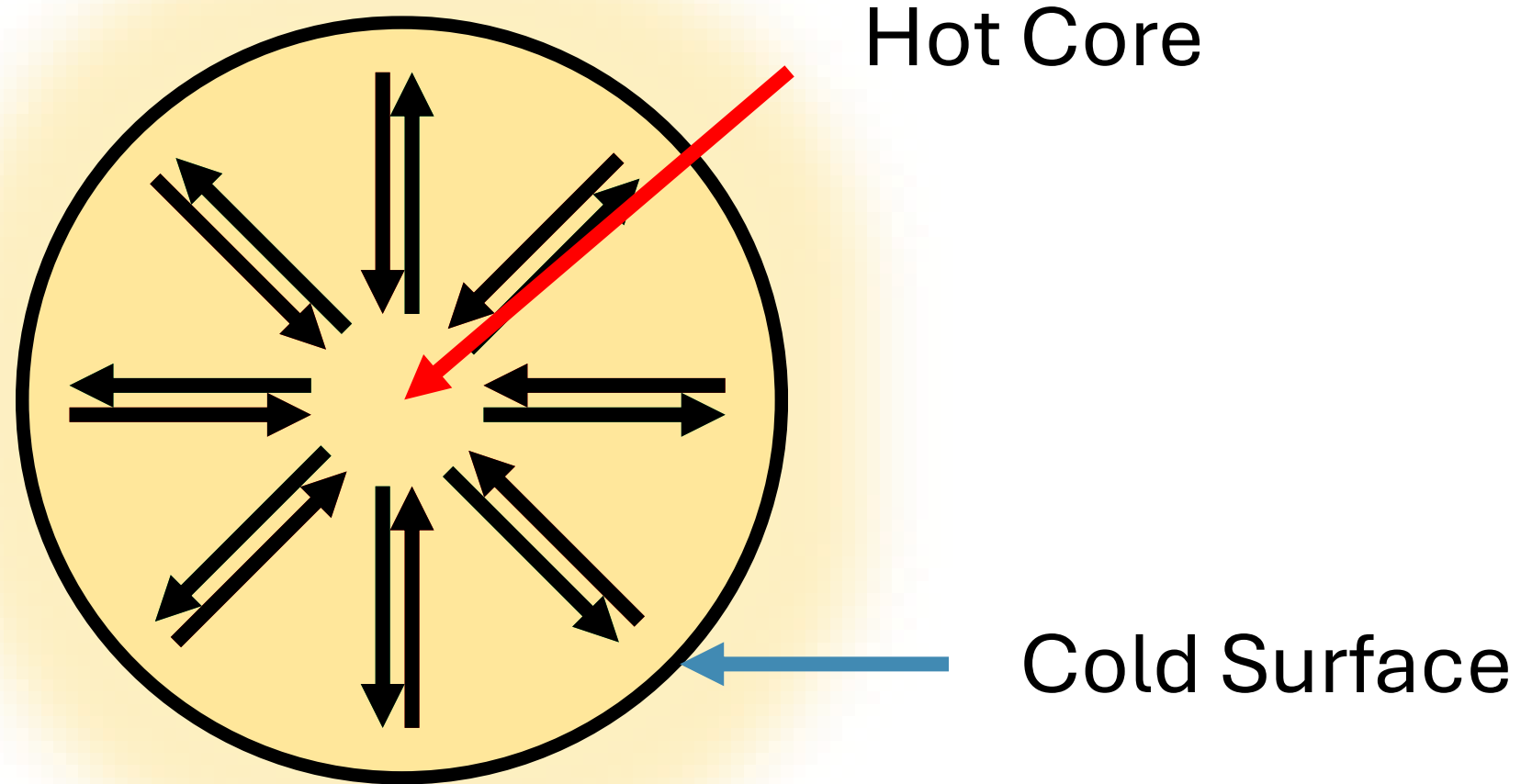
Stellar Interiors



Gravity acts
inwards

Fusion occurs in
the core

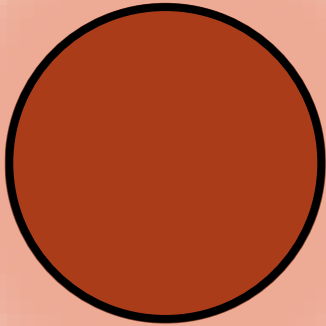
Pressure (gas +
radiation) which
acts outwards



Stellar Interiors

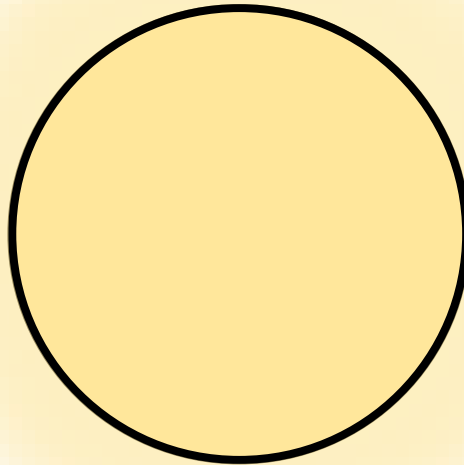


Low Mass



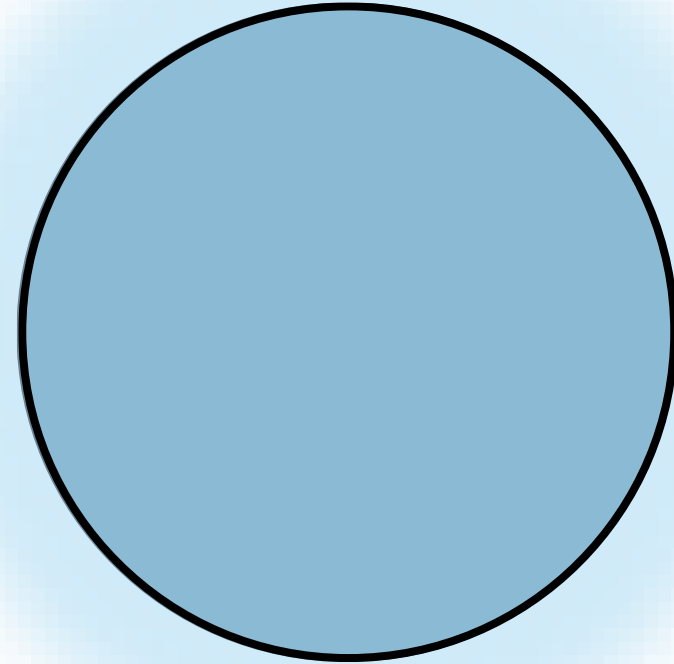
Fully Convective

Mid Mass



Convective Envelope

High Mass

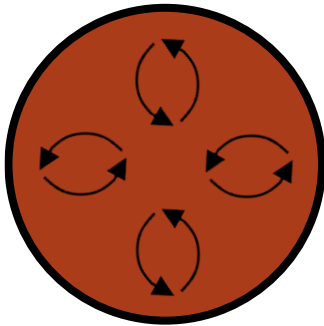


Convective Core

Stellar Interiors

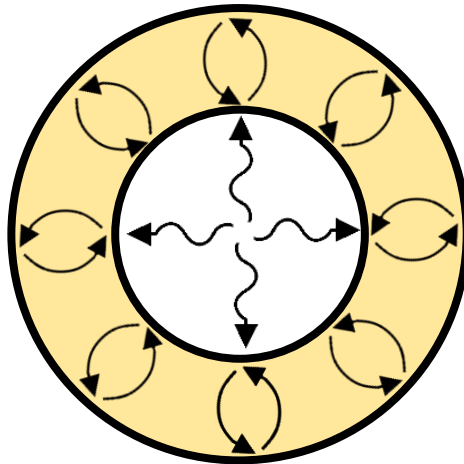


Low Mass



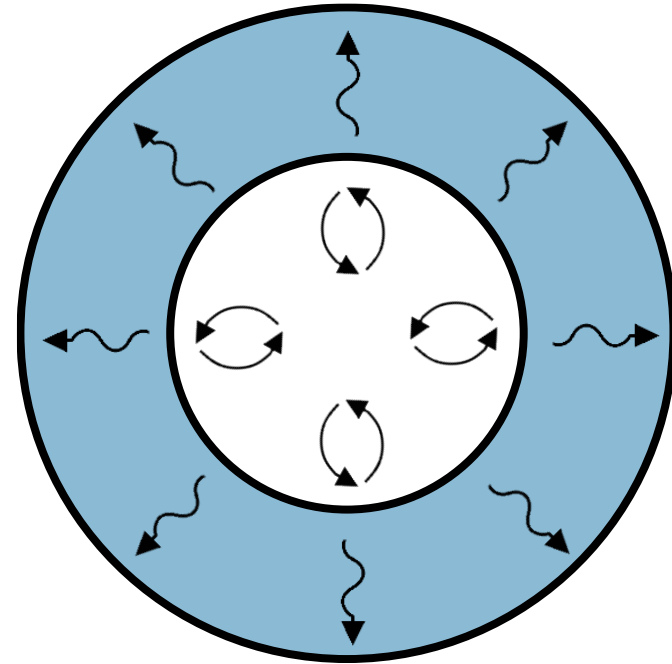
Fully Convective

Mid Mass



Convective Envelope

High Mass



Convective Core



Stellar Interiors

Schwarzschild criterion

For a homogenous composition

$$\nabla_{\text{rad}} < \nabla_{\text{ad}}$$

Radiative gradient
(only radiative energy transfer)

Adiabatic gradient
(no heat exchange)

Radiative Regions ✓

Convective Regions ✗



What is CBM?

Convective Boundary

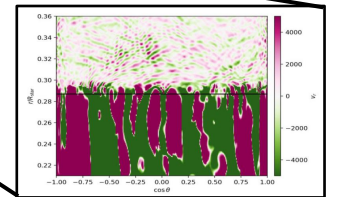
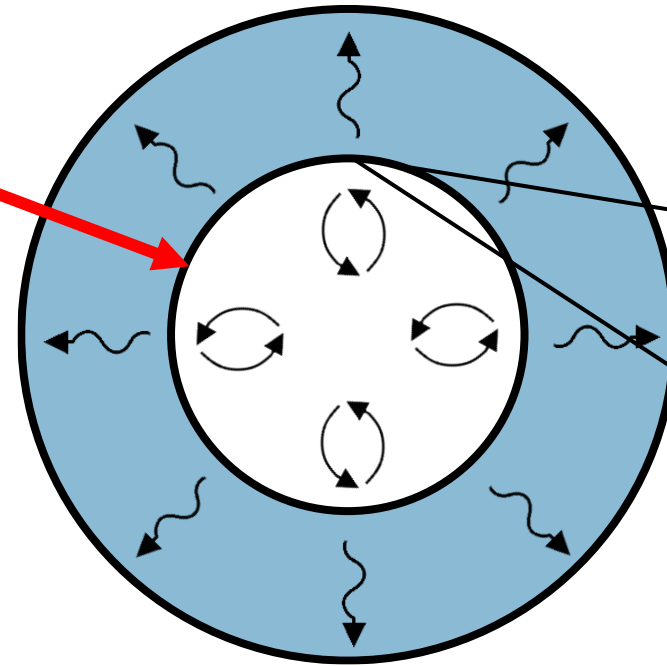
In radiative zone:

$$\nabla = \nabla_{\text{rad}}$$

In convective zone:

$$\nabla \approx \nabla_{\text{ad}}$$

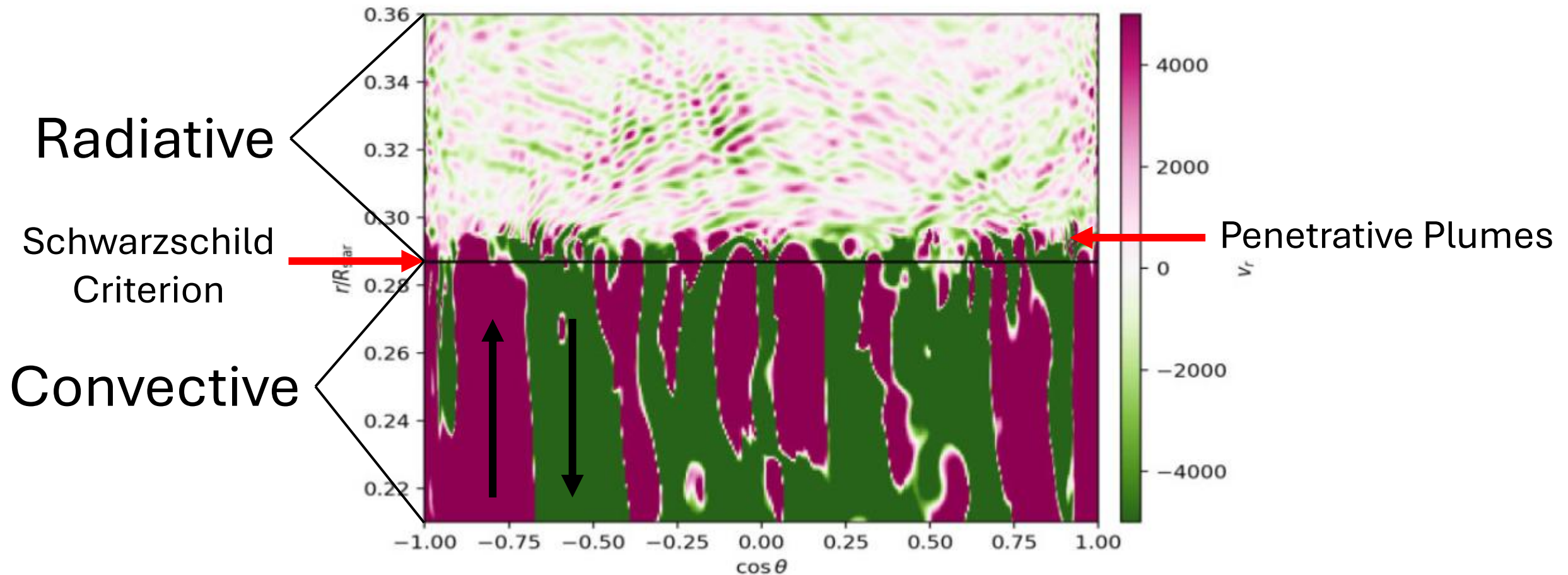
Boundary



What is CBM?



Penetration

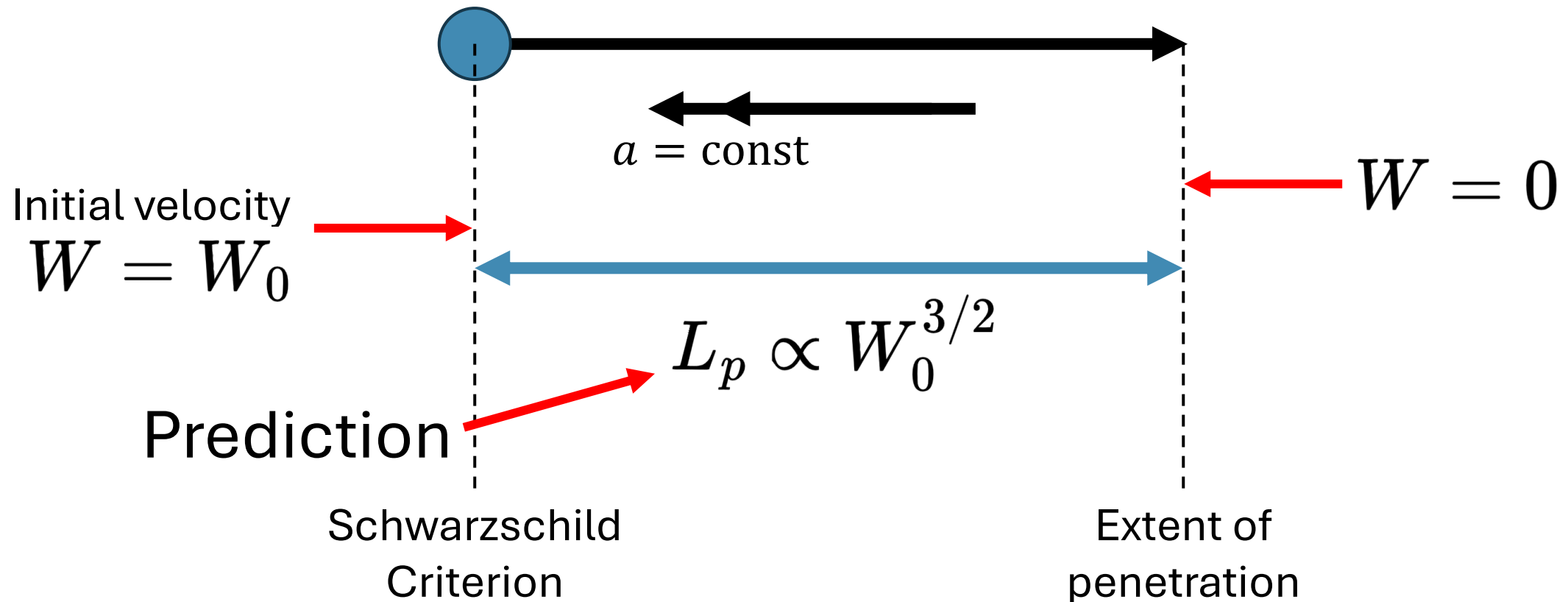




What is CBM?

Zahn theory

Treats as a ballistic problem (constant deceleration)



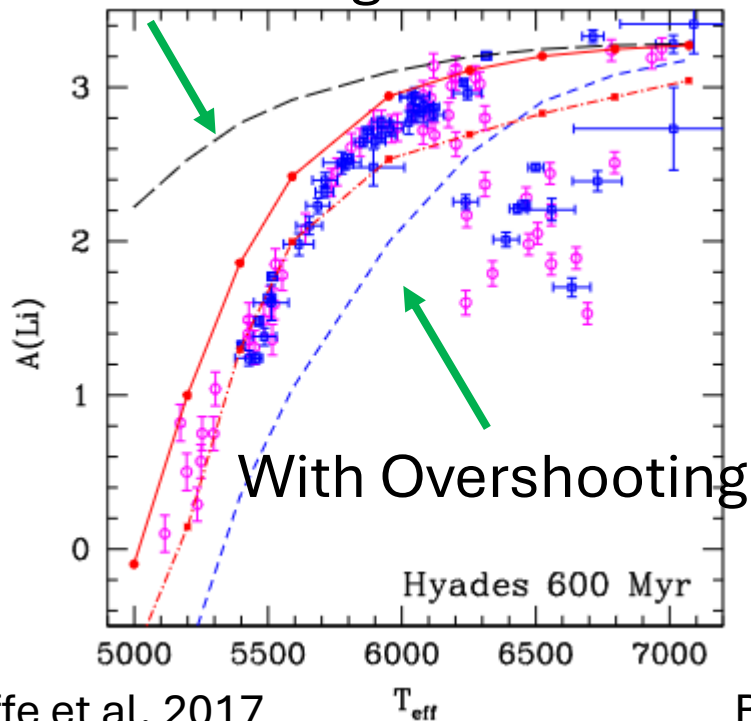


Observations

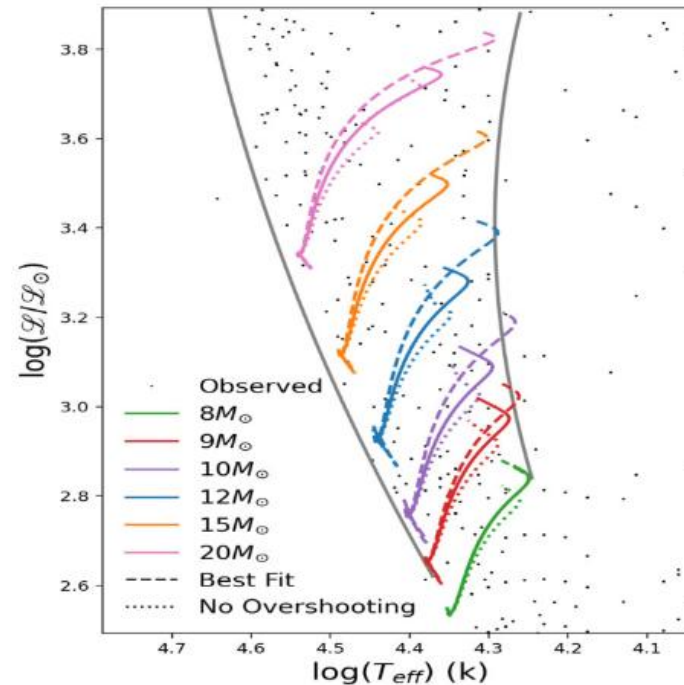
Evidence of penetration

Lithium depletion

No Overshooting

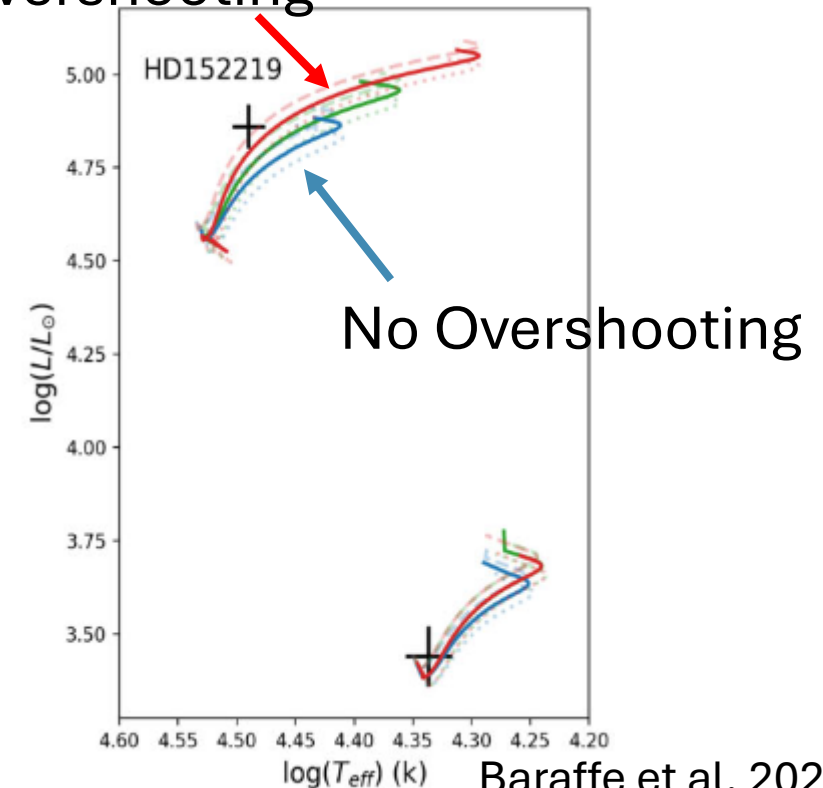


HR diagram

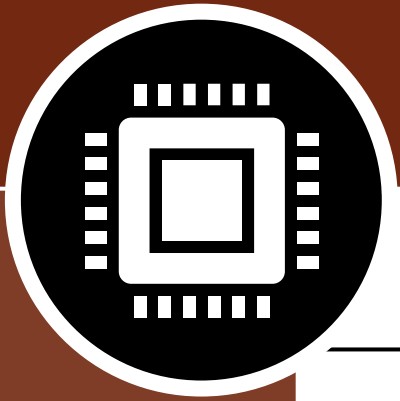


Massive binaries

With Overshooting



Stellar Simulations



Requirements

1D

2D

3D

Low

Medium

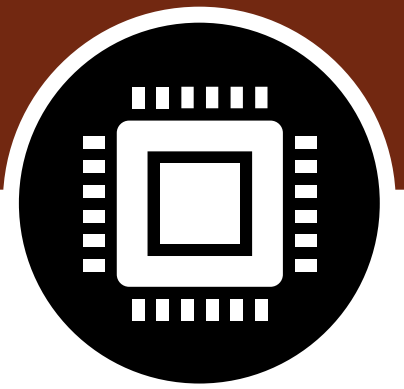
High

Evolutionary
models

Convection

Rotational
effects
+
Convection

Uses/Effects



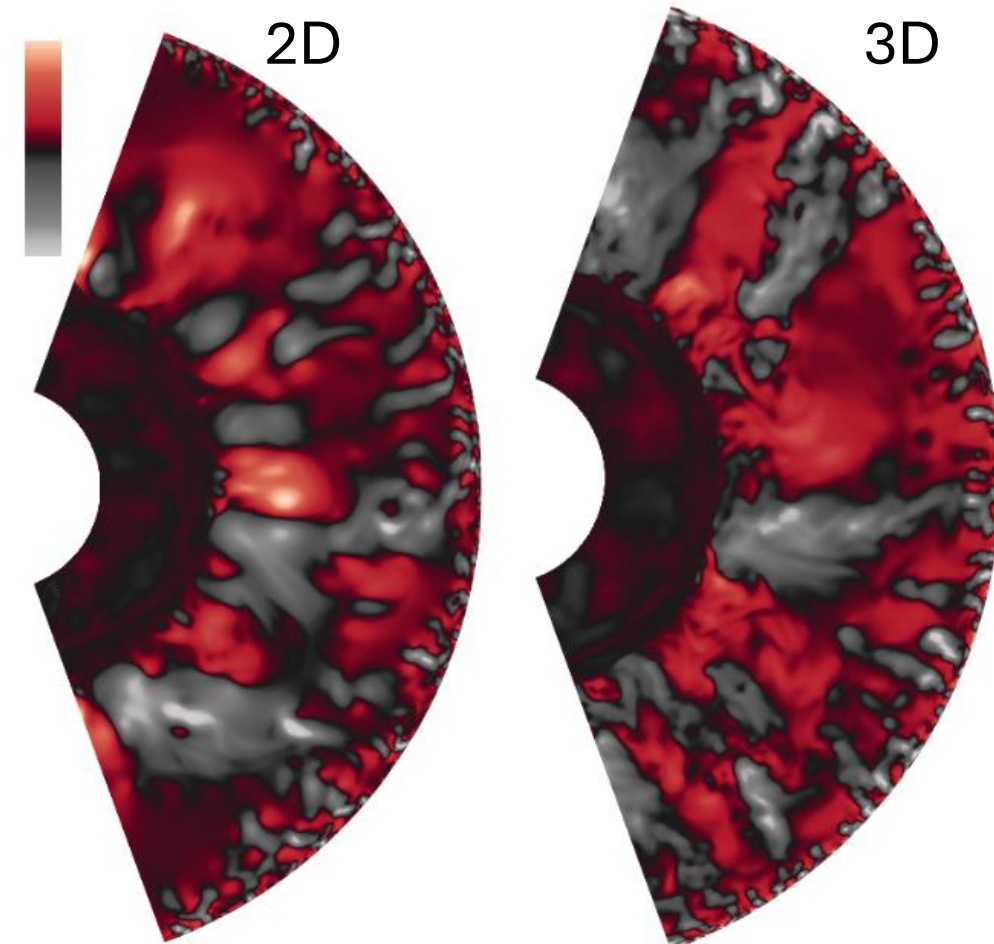
Stellar Simulations

321D link

The idea of using higher dimension simulations to inform and improve the lower (faster) models

3D→2D : Rotation
(Baraffe et al. 2017)

2D→1D : Diffusion coefficient
(Pratt et al. 2017)

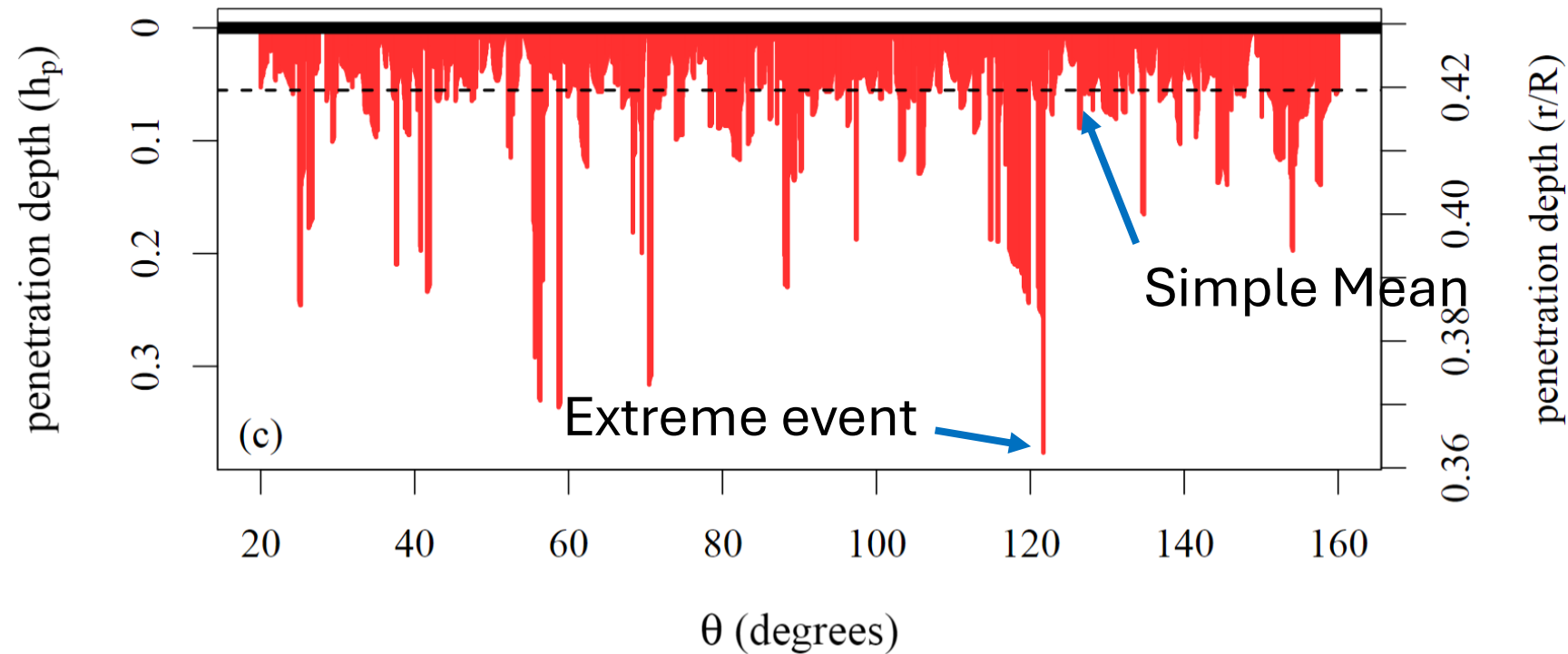




What has been done?

Extreme value theory

The effect of penetration is dominated by large infrequent plumes & follows Extreme value theory (EVT) (Pratt et al. 2017)



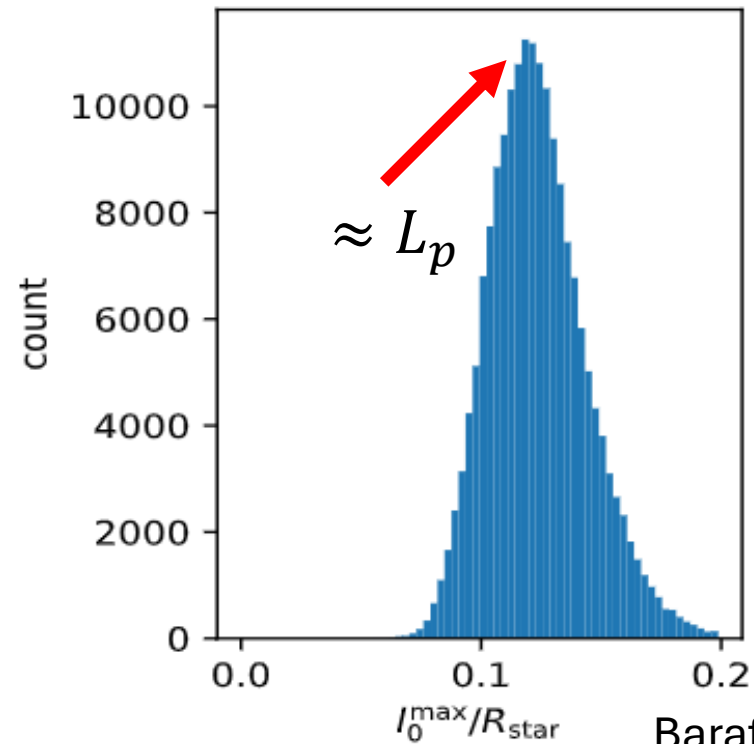


What has been done?

Extreme value theory

The effect of penetration is dominated by large infrequent plumes & follows Extreme value theory (EVT) (Pratt et al. 2017)

The statistics of EVT can be simplified by using the mean the maximum penetrations (Pratt et al. 2017, 2020)

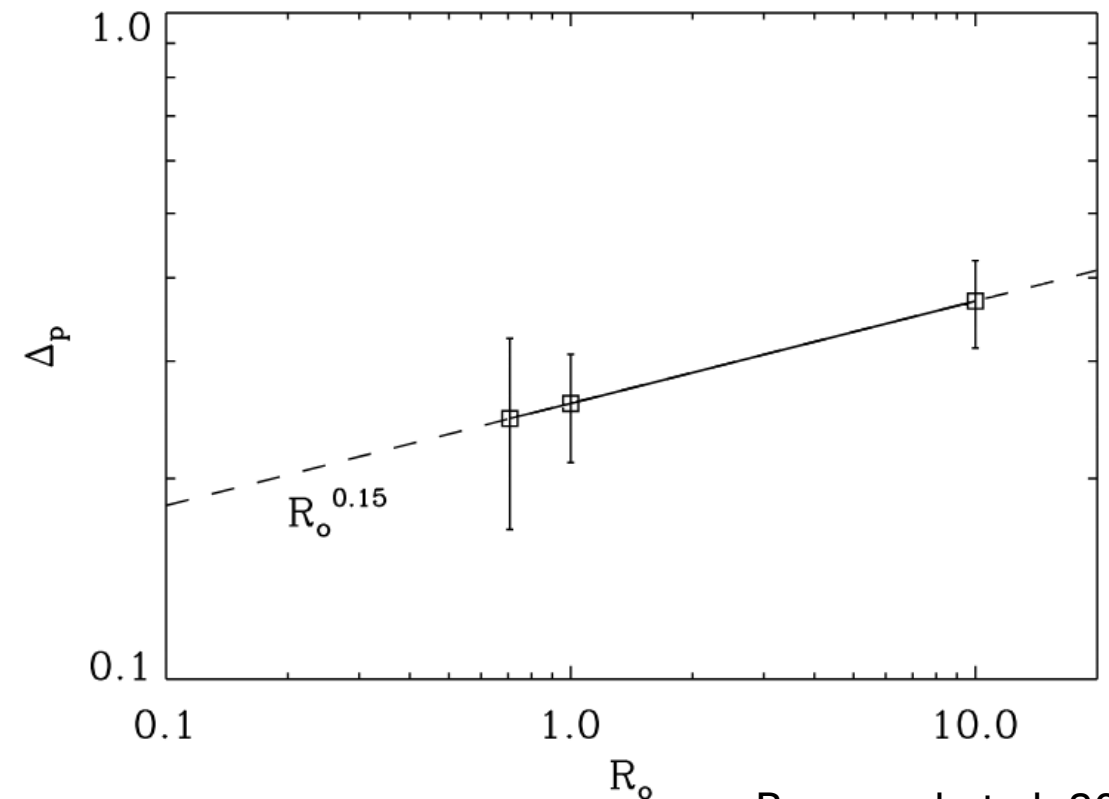




What has been done?

Rotational effects

Increased rotation leads to less penetration
(Brummell 2002, 2007)



Brummel et al. 2002

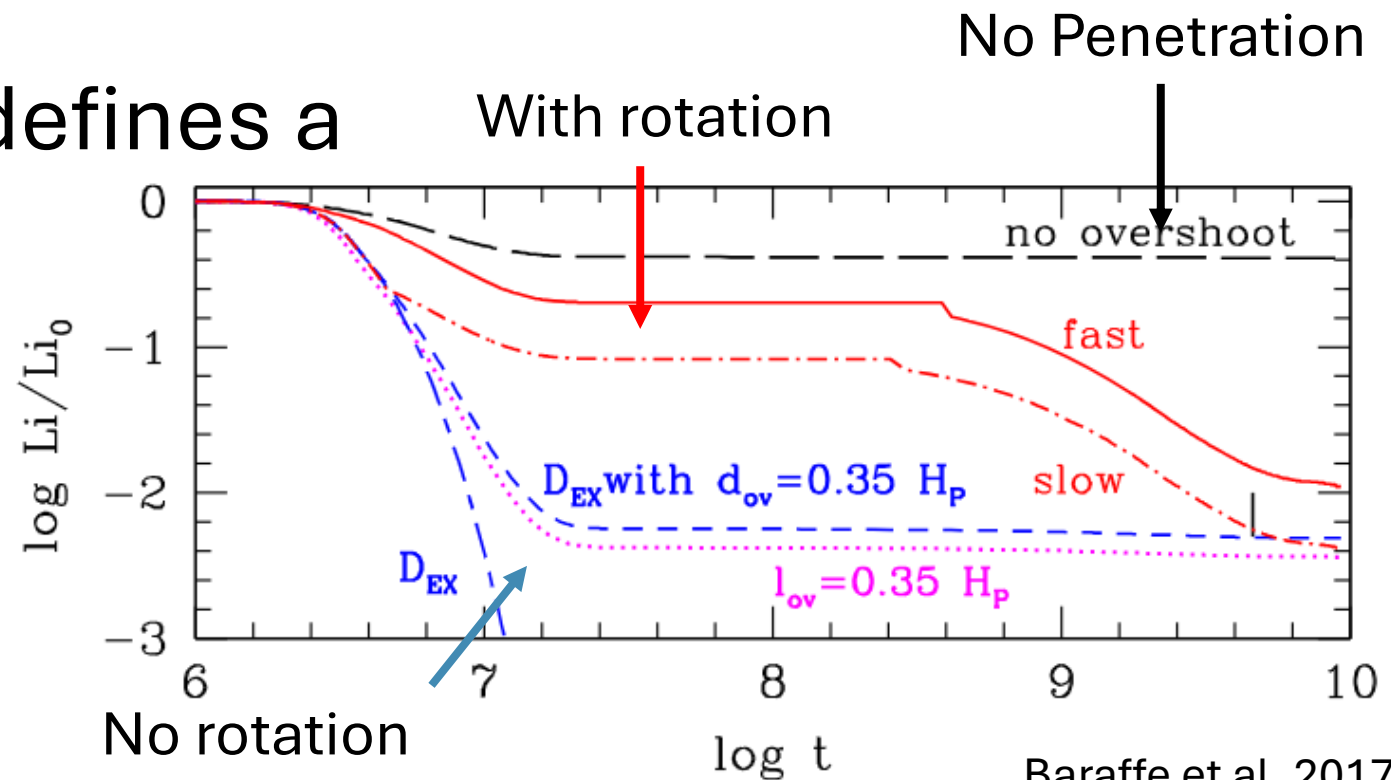


What has been done?

Rotational effects

Increased rotation leads to less penetration
(Brumell 2002, 2007)

To match Li observations defines a
critical rotation rate, $5\Omega_{\odot}$
(Baraffe 2017)





What has been done?

Rotational effects

Increased rotation leads to less penetration
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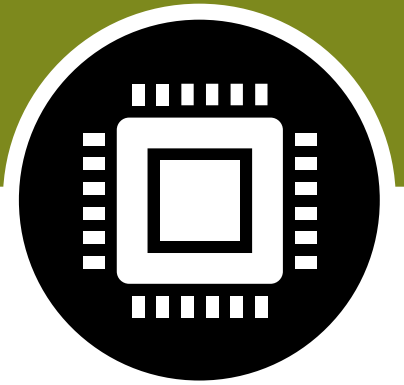
To match Li observations defines a
critical rotation rate, $5\Omega_{\odot}$

(Baraffe 2017)

Excited waves may cause penetration to
increase instead

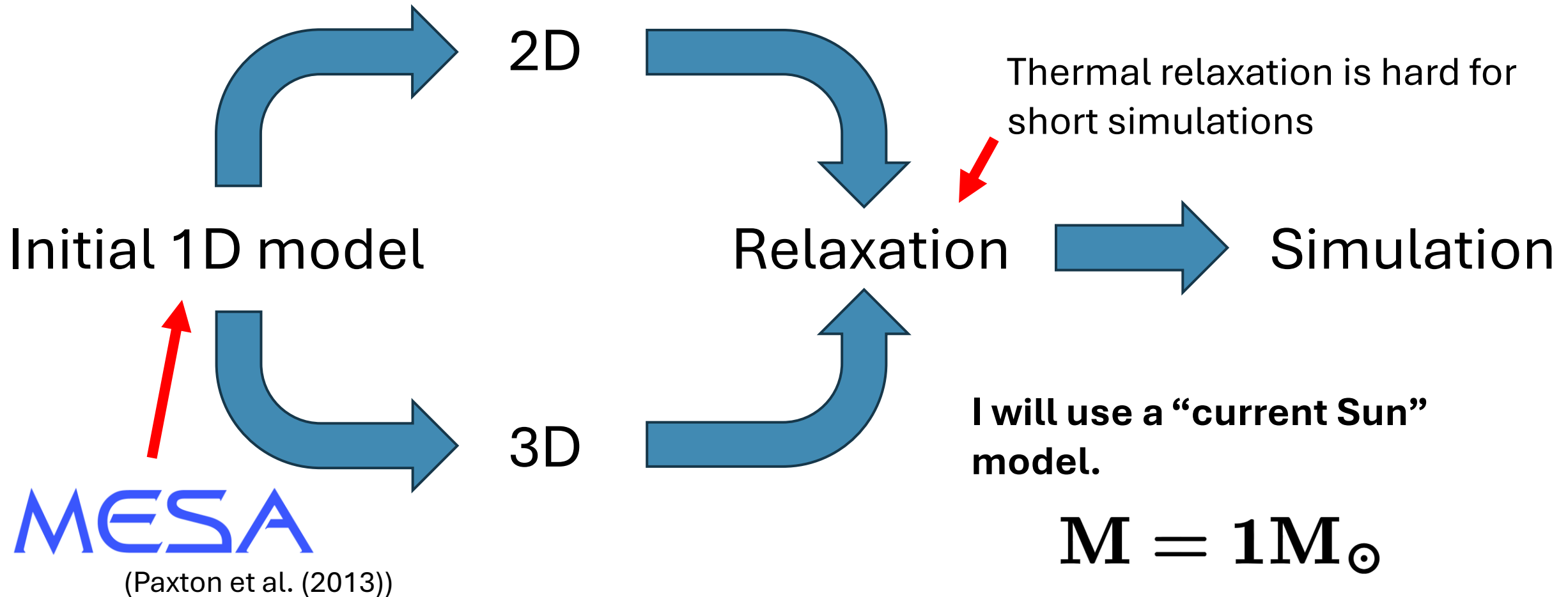
(Brun 2017)

What will I do?



Stellar Model - MUSIC

(Viallet, Baraffe & Walder (2011), Viallet et al. (2016), and Goffrey et al. (2017))



What will I do?

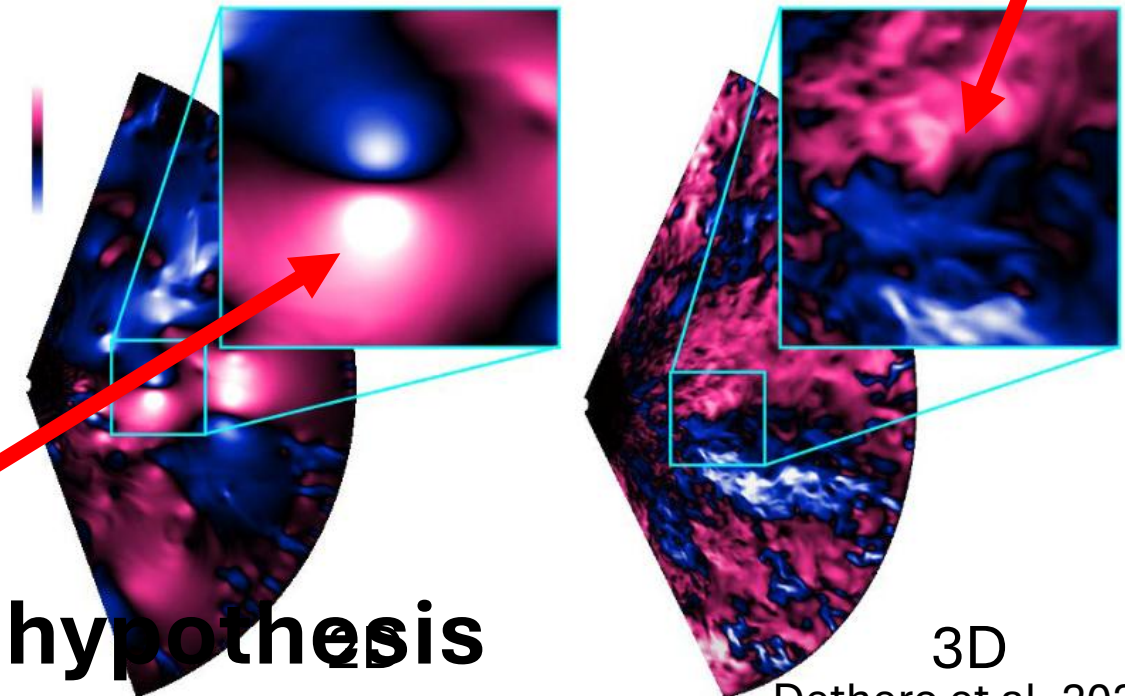


3D & 2D comparison

- 2D simulations give artificially higher velocities (e.g. Meakin & Arnett 2007, Arnett & Meakin 2011, Pratt 2020, Dethero 2024)
- Pratt (2020) suggests this is more on radial velocities and as such plumes

Typical velocity fields
Region very high speeds

I will test this hypothesis



3D
Dethero et al. 2024



What will I do?

Effect of Rotation on overshooting length

- Investigate, rotation leading to less overshooting

$\Omega/\Omega_{\odot}$: 0

Compare to
already existing
simulations
(simple case)

1

Compare to
Solar
observations

5

Identify trend,
and/or find
critical value



Summary

Outlook

Short Term:

- Start working with MUSIC
- Visualise velocity fields for 2D & 3D stationary case

Long Term:

- Compare 2D & 3D simulations
- Analyse Rotations effect on penetration

Thank you

Summary



References

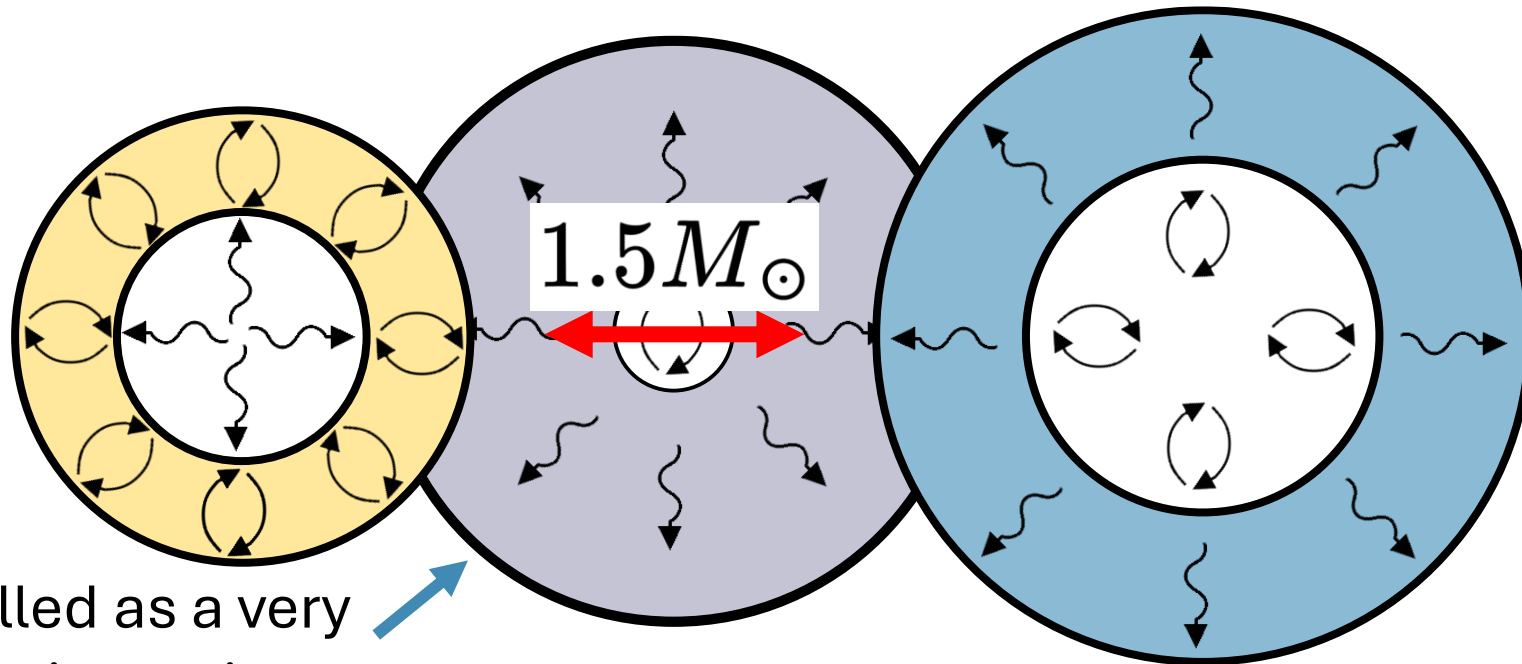
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Stellar Interiors

Mid mass to High mass switch

Switch because of dominant fusion reaction changes from pp to CNO (which creates a steep temperature gradient)



Usually modelled as a very small convective region



Stellar Interiors

Ledoux criterion

Similar to Schwarzschild but no assumption of homogeneous composition

density change with composition

$$\nabla_{\text{rad}} < \nabla_{\text{ad}} + \frac{\varphi}{\delta} \nabla \mu$$

Same as Schwarzschild

density change with temperature

chemical gradient



What has been done?

Baraffe 2017 - Rotational history

Assumes a solid-body rotation and magnetic braking using Kawaler's law (Kawaler 1988, also discussed in Viallet & Baraffe 2012)

